Estimating Active Dimension of Training Dynamics from One Scalar Log, with an Application to Grokking

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Abstract

Over a stretch of training in which the update rule is fixed and the parameters keep returning near states already visited, the states they revisit form a set of dimension far below the number of parameters: the active dimension of that regime. Measuring it needs the stored trajectory, whereas a run already writes scalar logs: how much of it does one such scalar carry? We reconstruct a state space by delay coordinates and estimate its intrinsic dimension by maximum likelihood on nearest-neighbour distances. The procedure returns a plausible number on any record, including one in which nothing recurs, so it estimates the active dimension only under conditions we establish separately. On six constructed systems of known active dimension it is accurate to about one component, up to a ceiling near eight. Statistics computed from the log identify the records on which the value is not a dimension at all. They reject both standard grokking settings, so we store a compressed record of the trajectory instead. Its effective rank collapses at generalisation in the four runs trained with weight decay, then re-expands, the simplification that mechanistic accounts of grokking predict. A run that merely stops moving can collapse as deeply. Only the timing of the collapse and its reversal mark the transition. Re-run over a window matched to the transition, the estimate on an ordinary log falls at the same step in those four runs and in neither of the other two, more steeply than in randomised surrogates of each run.

1 Introduction

A network is trained in a parameter space of dimension P, but over a stretch of training the parameters usually move in far fewer dimensions. Is an apparent simplification of training a real reduction in the number of degrees of freedom in use, or an artefact of a shrinking scale? Can a phase transition such as delayed generalisation be detected without instrumenting the run? Both questions are argued with whichever count of degrees of freedom is least expensive to compute. The counts in use disagree: the *available* dimension of the subspace the parameters are confined to [Li et al., 2018], the *functional* dimension of the map from that subspace to the outputs, and the *effective rank* of the trajectory covariance.

Our two questions concern a fourth quantity: the number of independent components of the set the parameters occupy while the update rule is fixed. We call it the *active dimension* of that regime (equation (1)). If r independent quasiperiodic phases drive the motion, the path lies on an r-dimensional torus and the active dimension is r. Section 5.2 moves the active dimension while the available and functional dimensions stay fixed, and section 6.4 separates it from the effective rank.

Measuring the active dimension means storing the parameter trajectory, which an ordinary run does not do. A run already writes scalar logs: a loss, a gradient norm, a parameter norm. *How much of the active dimension can be recovered from one such scalar?* Nonlinear time series analysis supplies a standard answer: reconstruct a state space from the scalar by delay coordinates [Takens, 1981, Sauer

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

et al., 1991], then estimate the intrinsic dimension of the reconstruction by maximum likelihood on nearest-neighbour distances [Levina and Bickel, 2004, MacKay and Ghahramani, 2005].

The procedure returns a number on any record. Power-law noise yields a finite and reproducible value fixed by its spectral exponent [Osborne and Provenzale, 1989, Theiler, 1991]; a decaying transient yields a large value. Establishing which case a training log falls in is the substance of the problem.

Contributions. We evaluate the estimate against a constructed ground truth on six systems of increasing realism, ending with a linear head trained inside a k-dimensional subspace on top of a frozen image network (section 5); we establish the conditions its value depends on, among them the identifiability ratio and the trend-crossing count, which identify the failure regimes from the series alone (section 6); and we apply the procedure to grokking, where both published settings fall outside the regime the estimate needs and the result comes from the stored trajectory instead (section 7).

2 Related work

Reconstruction and intrinsic dimension. Delay reconstruction follows from Takens' theorem and its extensions (Appendix A), is used on empirical series [Sugihara and May, 1990, Sugihara et al., 2012], and takes its lag and dimension from mutual information, false nearest neighbours or singular spectra [Fraser and Swinney, 1986, Kennel et al., 1992, Cao, 1997, Broomhead and King, 1986]. We use pooled maximum likelihood [Levina and Bickel, 2004, MacKay and Ghahramani, 2005]; the alternatives [Grassberger and Procaccia, 1983, Facco et al., 2017, Camastra and Staiano, 2016] are bounded alike by a finite record [Eckmann and Ruelle, 1992]. Oversampling [Theiler, 1986] and power-law noise [Osborne and Provenzale, 1989] produce spurious dimensions; surrogate tests [Theiler et al., 1992, Schreiber and Schmitz, 1996] detect nonlinear structure but not how much is present and miss a decaying transient entirely.

Dimension in optimisation. Random-subspace training [Li et al., 2018, Aghajanyan et al., 2021] reports the smallest k that still solves a task, a property of the task and not of a window; we adopt the same parameterisation but measure a different quantity within it. Closest to the active dimension is the finding that descent occurs largely in a small subspace spanned by the top Hessian eigenvectors [Gur-Ari et al., 2018, Sagun et al., 2017]. Ansuini et al. [2019] and Pope et al. [2021] estimate the intrinsic dimension of a data set or of a layer's representations, which are properties of a point cloud, whereas the active dimension is a property of a path. The participation ratio we adopt [Gao et al., 2017] is not the spectral-entropy quantity of the same name [Roy and Vetterli, 2007]; it has been measured on a grokking trajectory once, in passing [Notsawo Jr. et al., 2023]. All of these quantities need weights, gradients or activations, whereas the nearest precedent for a scalar computed during training and acted on is the gradient noise scale [McCandlish et al., 2018]. Mini-batch descent near a minimum is modelled as a stochastic process [Mandt et al., 2017].

Grokking. Delayed generalisation [Power et al., 2022] is studied in two settings and we use both. The transformer setting is regularised and, in our version, stochastic: the configuration of Nanda et al. [2023], also used by Liu et al. [2023], on modular addition and on the S_5 task of Power et al. [2022]. The perceptron setting is the unregularised full-batch gradient descent of Gromov [2023], on the modular polynomials and representability criterion of Doshi et al. [2024], which supply label-matched pairs (Appendix J). Explanations are framed in weight or activation space [Nanda et al., 2023, Varma et al., 2023, Liu et al., 2022, Barak et al., 2022] or in a lazy-to-rich transition [Kumar et al., 2024]. Closest to us, Notsawo Jr. et al. [2023] predict grokking from early loss oscillations [Thilak et al., 2022] and Merrill et al. [2023] track competing subnetworks. We differ from all of these in estimating a dynamical quantity whose ground truth is measured independently.

3 Problem statement

3.1 The active dimension

Let $\boldsymbol{s}_t = (\boldsymbol{\theta}_t, \boldsymbol{u}_t)$ be the optimiser state at step t: the parameter vector $\boldsymbol{\theta}_t \in \mathbb{R}^P$ and whatever the optimiser carries beside it, two accumulated moments under AdamW and nothing under plain gradient descent. Where the parameters are confined by construction or by the dynamics to an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{V}^\top \boldsymbol{c}$ with $\boldsymbol{V} \in \mathbb{R}^{k\times P}$ orthonormal, we work in the free coordinates $\boldsymbol{c}_t \in \mathbb{R}^k$; that change of coordinates is an isometry and preserves intrinsic dimension.

We call k the *available* dimension. The *functional* dimension is the rank of $\partial \boldsymbol{f}(\boldsymbol{c})/\partial \boldsymbol{c}$, with $\boldsymbol{f}$ the outputs on a fixed probe set; it bounds how many of the k directions can change the computed function. The *active* dimension, the subject of this paper, is a property of a *regime* R: a stretch of training over which the update rule is fixed and the state carries an ergodic invariant measure. Carried to the free coordinates, that measure gives the *occupation measure* μ_R, the long-run fraction of steps at which $\boldsymbol{c}_t$ falls in a given set. The regime occupies the support $\mathcal{A}_R$ of μ_R, the points whose every neighbourhood is visited with positive frequency. The active dimension is the exponent with which μ_R fills a small ball about such a point,

$$d_{\mathrm{act}}(R) \;=\; \lim_{\varepsilon\downarrow 0} \frac{\log \mu_R\big(B(\boldsymbol{c},\varepsilon)\big)}{\log \varepsilon}, \tag{1}$$

with $B(\boldsymbol{c},\varepsilon)$ the ball of radius ε about $\boldsymbol{c}$. This defines $d_{\mathrm{act}}(R)$ when the limit is the same at μ_R-almost every $\boldsymbol{c}$. Where $\mathcal{A}_R$ is a smooth d-dimensional manifold carrying a positive density the exponent is d; a point of it is then fixed by d local coordinates, the *active components*. Another chart supplies different ones, of the same number.

A window $\mathcal{W}$ of L consecutive optimiser steps holds L points, whose own exponent is zero. A window is therefore a sample from μ_R rather than the set itself. It estimates $d_{\mathrm{act}}(R)$ only where it lies inside one regime and only over the radii its neighbour distances span (equation (6)): a phase of smaller amplitude than those goes unresolved (section 6.4).

The systems of section 5 set $\boldsymbol{c}_t = F(\varphi_1(t), \ldots, \varphi_r(t))$ with r rationally independent quasiperiodic phases and F a smooth embedding of the r-torus, giving $\mathcal{A}_R = F(\mathbb{T}^r)$ and $d_{\mathrm{act}}(R) = r$. Rational independence makes the orbit dense on the torus; the embedding keeps F from merging or discarding a phase.

We also measure the participation ratio [Gao et al., 2017], which for a non-negative spectrum $\lambda_1, \ldots, \lambda_n$ is

$$\mathrm{PR}(\lambda) \;=\; \frac{\big(\sum_i \lambda_i\big)^2}{\sum_i \lambda_i^2}. \tag{2}$$

It equals m exactly when m of the λ_i are equal and the rest vanish. Applied over a window to the covariance of $\boldsymbol{c}_t$ it gives the *effective rank* $\mathrm{PR}^{\mathrm{pos}}$ and to the covariance of the residual $\tilde{\boldsymbol{c}}_t$ about the window's least-squares line the detrended form $\mathrm{PR}^{\mathrm{det}}$. A drifting trajectory needs the detrended form, since a straight drift dominates the spectrum and holds $\mathrm{PR}^{\mathrm{pos}}$ near unity however the trajectory moves about that line. Written without a superscript, *the effective rank* means $\mathrm{PR}^{\mathrm{pos}}$.

The effective rank is not the active dimension. It equals r when r directions carry comparable variance and falls below r when they do not, while $d_{\mathrm{act}}(R)$ stays at r (section 6.4). We use it to check that each system of section 5 really does excite r independent phases (table 11); the ground truth is the phase construction, not this measurement. It is also the quantity section 7.2 measures on a stored grokking trajectory.

3.2 Observation model and estimator

We observe a scalar series $x_t = \phi(\boldsymbol{s}_t)$, in which $\boldsymbol{s}_t$ is the full optimiser state and ϕ is fixed before the run. From it we reconstruct $\boldsymbol{y}_t = (x_t, \ldots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant set of box-counting dimension d, this map is an embedding once $E > 2d$ (theorems 1 and 2). Genericity is not verifiable for a ϕ fixed in advance, so every result below is reported over a family of observers. We estimate the intrinsic dimension of $\{\boldsymbol{y}_t\}$ by maximum likelihood on nearest-neighbour distances (theorem 4). Let $r_1^{(i)} \le \cdots \le r_m^{(i)}$ be the distances from point i to its m nearest neighbours, where a neighbour counts only if it is more than W_T steps away from i in time. Discarding the near-in-time neighbours in this way is the *Theiler exclusion* [Theiler, 1986, Kantz and Schreiber, 2004]: without it, the nearest points in $\mathbb{R}^E$ are the samples just before and after i, so the statistic measures the sampling rate instead of the geometry. Pooling the likelihood over the N points before inverting, as MacKay and Ghahramani [2005] recommend, gives

$$\hat{d}_{\mathrm{MG}}^{-1} \;=\; \frac{1}{N(m-1)} \sum_{i=1}^{N} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}, \tag{3}$$

The estimate is the value MG returns; *the estimator* is the procedure. Appendix B.1 gives the algorithm as implemented; table 8 scores the per-point average of Levina and Bickel [2004] beside it. The *roughness* $\operatorname{std}(\Delta x)/\operatorname{std}(x)$, the *linear participation ratio* $\mathrm{PR}_{\mathrm{delay}}$ of equation (2) on $\operatorname{Cov}\{\boldsymbol{y}_t\}$ and the *spectral participation ratio* on the power spectrum of x accompany every estimate. Each can grow with r for reasons that involve no geometry. If one of them recovers the truth as well as the estimator does, the accuracy of the estimator is no evidence that a dimension has been measured.

3.3 Dynamical regimes

Two conditions must hold before an estimate computed from a log gives an active dimension. The embedding theorems (theorems 1 and 2) require a deterministic map on a compact invariant set; that licenses the delay reconstruction as a change of coordinates. The neighbour statistic of equation (3) requires more: the record must cover that set densely enough that the Theiler exclusion still leaves each point a genuine return. The estimator itself would accept points drawn from independent runs; recurrence is needed because our entire sample is one trajectory.

- **Deterministic and recurrent.** The path returns close to states already passed through and lies on an r-dimensional torus, so $\mathcal{A}_R$ has r independent components and $d_{\mathrm{act}}(R) = r$. Here both conditions hold and the estimator recovers r.
- **Deterministic and transient.** A converging run decays along a curve it never revisits. The curve itself is one-dimensional, but a run that does not return samples no invariant measure, so it has no $\mathcal{A}_R$ and equation (1) has nothing to measure. The dimension of a path and the dimension of a regime are different quantities. The value returned here is invalid rather than a dimension of one (section 6.1). The effective rank does stay near unity (table 11), but only because the curve is nearly straight.
- **Stochastically driven.** A stationary mini-batch optimiser may have an occupation measure of its own, whose dimension counts the directions the gradient noise excites. That is not the quantity we are after. The embedding theorems for forced systems (theorem 3) do cover a drive of this kind, but one realisation of it at a time, whereas a single run crosses realisations.

3.4 Diagnostics

The series alone is enough to tell a transient from a stochastic drive. The **identifiability ratio** $\rho_{\mathrm{ident}} = \hat{d}_{\mathrm{MG}}(2E_{\max})/\hat{d}_{\mathrm{MG}}(E_{\max})$ compares the estimate at a doubled embedding dimension with the estimate at the working one. It stays near unity where a dimension can be measured. Under mini-batch noise the estimate follows the embedding dimension rather than the data; the ratio then grows. A transient escapes it: there the estimate is stable in the embedding dimension for reasons unconnected with any dimension. A transient is caught by the **trend-crossing count** instead, the number of sign changes of the residual of the window about its own least-squares line, which is near zero on a monotone decay. The count tests only non-monotonicity, so it cannot replace the ratio; we report both with every estimate.

4 Validation protocol

The requirements below were fixed before any result was computed; table 4 records which of them each system fails.

Constructed truth: the ground truth follows from the construction rather than from a fit. Every system is built with r independent quasiperiodic phases, so that $d_{\mathrm{act}}(R) = r$; the effective rank of the recorded trajectory must then confirm that all r are comparably excited.

Frozen configuration: the estimator is chosen once, by a search over its own settings alone. Searching the settings of the system at the same time would select the data together with the method: the result would then report how favourable a system the search had found.

Withheld ranks: ranks are held out as well as seeds, because the geometry of the excitation depends on r and every seed at a given rank shares that geometry.

Zero learning rate: every system is re-run with the learning rate set to zero, so that the parameters never move and any observer of the optimiser must be constant. A log that still varies and still returns its earlier estimate was never a function of the optimiser.

5 Evaluation on systems with a known active dimension

We evaluate the estimator on six systems of increasing realism, from an oscillating diagonal matrix to a linear head on a network trained on image data (table 4). In each the measured effective rank confirms that all r phases are excited; the estimator receives one scalar series and nothing else.

5.1 An oscillating diagonal matrix

The first system has no optimiser and no feedback. In $\boldsymbol{W}(t) = \mathrm{diag}(b_i + \delta_i(t))$, the offsets b_i are fixed and r of the δ_i oscillate at rationally independent frequencies. The estimate tracks r across the whole withheld range to within a third of a component (table 4).

5.2 A constrained head on a network trained on image data

A tanh network is trained on the scikit-learn handwritten digits and then frozen. A linear head confined to a k-dimensional affine subspace is trained on top of it under softmax cross-entropy, so that the curvature is data-dependent and all the dynamics occur in $\boldsymbol{c}$. The excitation is made r-dimensional by modulating the loss weights of r of twelve fixed data groups (Appendix E). We drive the head quasiperiodically, with mini-batch noise, and by plain gradient descent, covering the recurrent, stochastic and transient regimes of section 3.3.

Twelve scalar observers are recorded (Appendix B.2). The estimator was frozen on four of them, on seeds and ranks reserved for that purpose, so every result below is out of sample in rank, in seed, and for most observers in the observer too. In the deterministic recurrent regime the estimate recovers the active dimension to within about one component. Every observer orders the withheld ranks correctly (table 4); for no scored observer does the roughness of the log order them, so the agreement cannot be explained by the logs merely becoming rougher as r grows (table 6).

The ceiling. The estimator saturates above about eight components. At $k = 20$ it tracks the rank to about ten and then falls behind (table 7). The largest r it still tracks follows neither the embedding dimension nor the record length and never exceeds about eleven (Appendix L). One caveat: the participation ratio of the delay covariance, a purely linear statistic, saturates along the same curve, so the limit may be the number of components the delay window resolves linearly, with no geometry involved.

The zero-learning-rate test. Most published analyses of dimension on training logs do not run the check of section 4. It invalidated four of our six systems and the function-subspace variant besides, each of which returns its reported estimate with the parameters frozen, because the drive keeps the residual moving while the parameters do not. It also rejects the instantaneous mini-batch loss of this system, otherwise among the more accurate observers.

6 Conditions of validity

6.1 Recovery in each regime

In figure 1(a) the fast quasiperiodic drive follows the diagonal over the whole withheld range. Under mini-batch noise the estimate is flat in r and well above the truth, responding to $E_{\max}$ and to the length of the record instead of to the data. Weak noise on an otherwise recurrent trajectory leaves the ordering intact and the absolute value wrong. The slow drive is the informative case, because it changes behaviour partway along: it tracks the truth to about $r = 4$, then saturates.

Panel (b) plots the identifiability ratio on either side of that change. The regime labels alone would have predicted the extreme cases, not the slow drive, which passes at $r = 2$, where panel (a) puts the estimate near the truth, and fails at $r = 6$, where panel (a) puts it at little more than half the truth. The ratio therefore locates the failure inside a single system, using nothing the ground truth supplied.

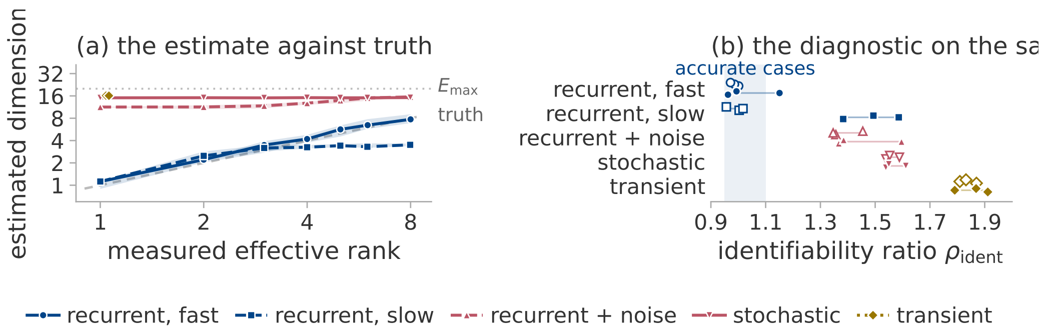

Figure 1: Recovery of the active dimension in each regime. The linear head of section 5.2, driven five ways: *recurrent, fast* and *recurrent, slow* quasiperiodic drives, *recurrent + noise* adding gradient noise to the fast one, *stochastic* mini-batch noise alone, and *transient* plain gradient descent. Colour is the regime of section 3.3. (a) Estimate against truth: marker, median over seeds and observers; band, their interquartile range; dashed diagonal, exact recovery; dotted line, $E_{\max}$. (b) The identifiability ratio on the same five, one point per observer, at $r = 2$ (open) and $r = 6$ (solid).

The Theiler exclusion is the reason the estimator errs most on a transient. There it returns about 29, a divergent quantity evaluated at the cap on the exclusion. With the exclusion removed, it returns at every rank, seed and observer the value maximum likelihood gives on a uniformly sampled curve, while the fast recurrent case is unchanged (Appendix I.2).

Every system evaluated above is externally driven, whereas training is not, so the evaluation establishes only a conditional result. The full-batch runs of section 7 do reach the edge of stability [Cohen et al., 2021], where an undriven run might still recur, but oscillation is not recurrence and the diagnostics reject every one of them (Appendix K).

6.2 The delay lag

The delay window spans $(E_{\max} - 1)\tau$ samples. Below an appreciable fraction of the oscillation period the delay coordinates are nearly collinear and the torus never unfolds [Casdagli et al., 1991]. Varying τ alone on a fixed torus moves the estimate over more than an order of magnitude, through the truth only where the window covers a few tenths of a period (figure 5). The lag must therefore be set from the oscillation period, which a constructed system supplies and a training log does not. From the grokking logs we accordingly take changes and never a level (section 7).

6.3 Nuisance factors

At $r = 4$ we vary factors that leave r unchanged over five seeds and twelve observers, measuring each shift against the shift a real change of four components produces (table 9). A ramped observer gain moves the estimate by a component and a half, so a run whose parameter norm grows steadily will drift by that much at fixed r. This is the confound that matters most in section 7. Smoothness matters less: shifting the drive band by an octave changes the roughness sharply while moving the estimate by about half as much. Roughness tracks the estimate closely here and again in section 7.3; Appendix M holds each fixed in turn while the other moves.

6.4 The estimand is a dimension, not an effective rank

Every construction in section 5 drives its r phases at equal amplitude, which makes the active dimension and the effective rank of section 3.1 numerically equal. Where two candidate estimands agree, an experiment cannot say which of them the estimator reports. Driving phase l at amplitude q^l separates them: the path still lies on an r-dimensional torus, while the effective rank falls as q falls. The estimate follows the active dimension (figure 6), staying closer to r than to the effective rank and registering only the phases it can resolve.

The diagnostics. Across this grid the identifiability ratio again separates the accurate cases from the inaccurate ones without overlap, now within a single regime as well as between regimes. The ratio and the trend-crossing count are calibrated on these cases alone, so the band drawn in section 7 records where the accurate cases fell and carries no authority beyond them.

7 Application to grokking

We apply the procedure to the grokking settings of section 2: a transformer trained with weight decay and AdamW on mini-batches, which places it in the stochastic regime of section 3.3, and a two-layer perceptron trained by unregularised full-batch gradient descent (Appendix J lists every run).

7.1 The diagnostics reject both settings

We run the frozen configuration of section 5.2 on these logs, shortening the window and stride to fit the records and changing nothing else (table 12, figure 7). The regularised transformer runs are unstable in the embedding dimension, their estimates ranging from four to thirteen components across configurations. The perceptron runs and the transformer runs without weight decay carry the signature of a transient instead: two trend crossings in a whole record, at a ratio still far outside the band. Those undecayed runs are trained on mini-batches, so the regime label assigned in advance does not decide the outcome. The level is therefore not a dimension in either setting. A change within a single run could still be one, since a fixed configuration adds the same offset to every window of that run. This window cannot show it: it spans tens of thousands of optimiser steps against a transition a few hundred steps wide.

Label-matched pairs. The perceptron targets come in pairs differing by a single monomial, both members of which fit the training set completely under identical optimisation. The estimate does separate them, but every one of these runs is transient and the separation is already visible in the raw loss curves (Appendices F, H.1 and H.2).

7.2 The trajectory simplifies at generalisation

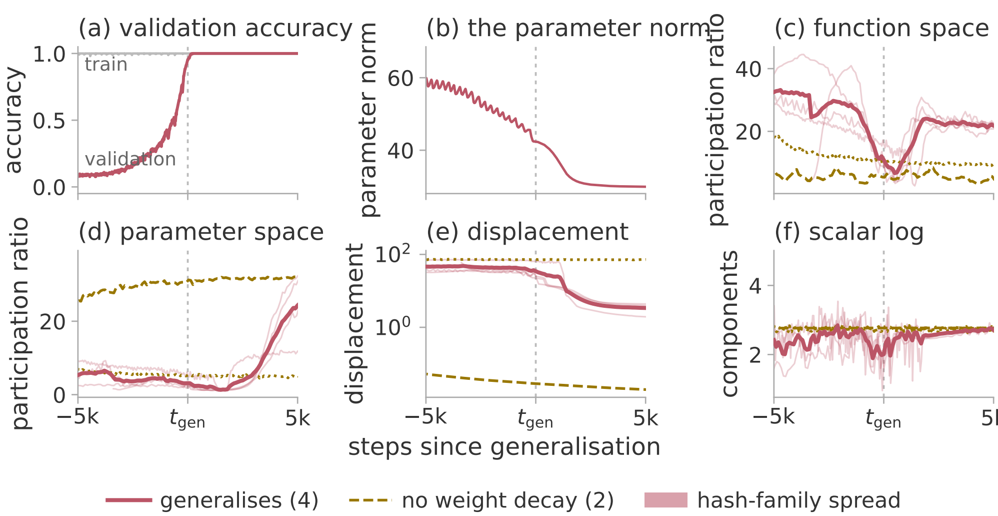

Figure 2: Simplification of the trajectory at generalisation. Three regularised seeds of modular addition and one S_5 composition, against the same configurations trained without weight decay, aligned on the generalisation step of the run each matches. Thick line, the mean of the four generalising runs; band, the spread between the hash families of Appendix G.1. Panels (a) and (b) are the one run whose training curve is stored; (c)–(e) plot $\mathrm{PR}^{\mathrm{det}}$ and the displacement over windows of 600 steps, 300 for S_5; (f) the estimate of section 7.3 on the log of (b).

The effective rank is defined whether or not the path returns on itself, so a stored trajectory allows a measurement that a log does not. We store the trajectory of a transformer trained to grok, compressed at every logged step by a random projection of the parameter vector and of the probe logits (Appendix G.1). Near generalisation the effective rank collapses in both spaces, reaching near unity in parameter space. In each space the minimum falls after the transition and within two thousand steps of it. Both then re-expand (figure 2, table 14). The collapse might follow training progress, but it occurs at generalisation in seeds whose generalisation steps differ several-fold and in a second task. It might record a trajectory that has come to a halt, but the total displacement reaches its floor thousands of steps later and never recovers, whereas the effective rank does recover. One caveat: the effective rank measures variation inside the window. Only that variation collapses; $d_{\mathrm{act}}(R)$ itself may be unchanged.

The runs without weight decay. The same two configurations trained without weight decay memorise and stay at chance accuracy. Within the transition window they decline much less than the generalising runs, but somewhere in training one of them declines just as far (table 14). Depth therefore does not separate the generalising runs from the memorising ones; the timing of the decline separates them, as does whether it reverses. A low effective rank is reached both by a run that has come to rest and by one about to generalise.

7.3 The log estimate falls where the trajectory does

The test section 7.1 could not perform becomes possible once the window is set in units of the transition, not as a fraction of the record. We re-run the estimator on the same logs, at the window length and midpoints the direct measurement uses, so that both statistics are sampled at the same instants. The frozen configuration does not fit a window that short, so we report every cell of a grid and take as headline the cell named by a rule blind to the outcome (Appendix H.3).

On the parameter norm the estimate halves at the generalisation step in all four generalising runs (figure 2f). That drop is among the largest any of the four ever shows, while the other two runs show nothing remarkable there; all but two of the usable grid cells separate the generalising runs from the other two (table 16). We also compare each run against randomised series built from it that keep its slow shape and its fluctuation spectrum while removing its fine structure. The fraction of those series falling as steeply as the run itself is smaller for every generalising run than for either of the other two (Appendix H.3).

The same event is thus recorded twice: once by storing the trajectory, once in a log a run writes anyway. The second record is the weaker. A window as short as the transition need not lie inside one regime, so the fall is a signal that tracks the transition and not a measurement of equation (1). Its observer is the parameter norm, for which section 6.3 leaves a smoothness explanation open, so the comparison against randomised series carries the claim alone. The roughness of these windows falls by a far larger factor than the estimate, which merely halves: a simpler statistic registers the change more strongly.

8 Conclusion

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the active components of the regime, to within about one component and up to about eight. Outside that regime it does not: a transient supplies no returns; mini-batch noise contributes directions of its own. The estimator is usable only alongside the identifiability ratio and the trend-crossing count, which place both grokking settings outside the regime it needs. Our result there comes from the stored trajectory, whose effective rank collapses at generalisation and then re-expands.

Limitations. The measurement needs a recurrent regime and a lag matched to a period no log reveals. Above about eight components it is only ordinal. The collapse comes from four runs against two, distinguished by weight decay, which also drives the parameter norm; it marks the transition only once the transition has happened. It does not reproduce in the full-batch setting. The drop at the matched window is close to the drop a ramped gain alone produces. The edge-of-stability evidence comes from one task (Appendices B.3, H.1, H.3 and K).

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Appendix outline

These appendices are supplementary. The body stands without them; each holds the data, the construction or the audit behind a claim made there.

Appendix A states, without proofs, the four published results the estimator is assembled from: the delay embedding theorem of Takens [1981], its extension to fractal sets, the version for systems driven by an external input, and the maximum likelihood estimator of an intrinsic dimension. It closes with the limits a finite record imposes.

Appendix B records the procedure as run: the algorithm and its constants, the scalar observers of section 5.2, and the configurations at which the estimator was frozen, with the grids they were selected on.

Appendix C scores section 5 in full, observer by observer, against every simpler statistic we could compute on the same data. Three of those comparisons go against the estimator.

Appendix D gives the three experiments section 6 summarises, on the delay lag, the nuisance factors and unequal drive amplitudes, and a fourth the body does not report: whether a real change in the active dimension is detected when one occurs.

Appendix E describes how the image-data system of section 5.2 is built in the three variants the paper scores, and measures how many independent directions each excites. The function-space variant falls short of its own requirement.

Appendix F gives the diagnostic values behind section 7.1 for every training log, and shows the label-matched pairs beside the raw loss curves that limit any claim of a separation between them.

Appendix G covers the measurement of section 7.2: the random projection that makes storing a whole trajectory affordable, the checks that validate it, the spectral guarantee those checks do not supply, and the collapse run by run.

Appendix H makes one argument three times: a statistic computed over a window cannot localise anything shorter than that window. The three parts make it for the full-batch trajectory, for a run whose estimate declines although it never generalises, and for the shortened window of section 7.3, where that result is also qualified.

Appendix I audits the Theiler exclusion, the one setting that fixes the value returned on a transient and the one at which this paper departs from its own protocol. It measures the cost of that departure and varies the exclusion from zero upwards.

Appendix J gives every run its task, architecture, length and outcome, and states where our grokking settings depart from the published configurations they are taken from.

Appendix K tests whether gradient descent at the edge of stability is deterministic and recurrent at once, as the estimate requires, and how much of that survives a coarser log.

Appendix L separates the level the estimator returns at high rank from the highest rank at which it still tracks the truth.

Appendix M moves each of the two while holding the other fixed. It also reports two findings that count against the estimator.

Appendix N states the compute used and the material to be released.

A The results behind the estimator

The results below are stated without proof and in the form the paper uses. Every condition they impose is an assumption about the run rather than a property we can verify from the log.

Throughout, f is the map advancing the system by one step, ϕ the scalar observation, and

$$\Phi_{f,\phi}(\boldsymbol{s}) \;=\; \big(\phi(\boldsymbol{s}),\, \phi(f(\boldsymbol{s})),\, \ldots,\, \phi(f^{E-1}(\boldsymbol{s}))\big) \;\in\; \mathbb{R}^E \tag{4}$$

the delay map.

Theorem 1 (Delay embedding; Takens, 1981)**.** *Let M be a compact manifold of dimension d and let $E > 2d$. For a generic pair (f, ϕ) with $f \in C^2(M, M)$ a diffeomorphism and $\phi \in C^2(M, \mathbb{R})$, the delay map equation (4) is an embedding of M into $\mathbb{R}^E$.*

An embedding is one-to-one and an immersion, so distances in $\mathbb{R}^E$ between reconstructed points are a faithful, though distorted, image of distances between states. The set they fill is the object the neighbour statistic below measures (figure 3). Theorem 1 requires a manifold and the sets a training run occupies need not be one; the extension below is the version the paper relies on.

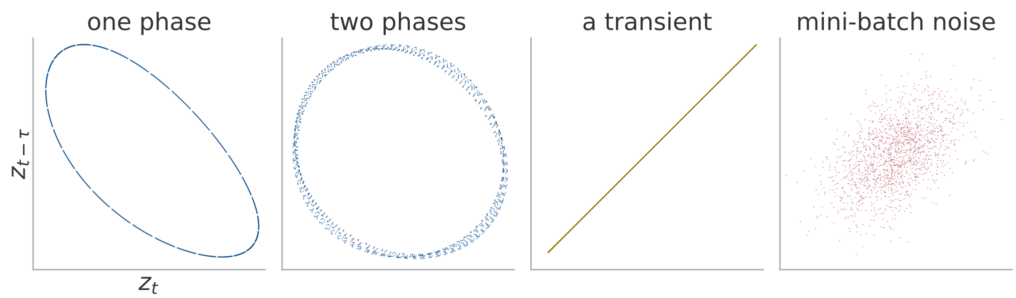

Figure 3: The set the neighbour search measures, in the plane of the first two delay coordinates. One window of the image-data system of section 5.2 under each excitation, reconstructed at the frozen configuration. A single phase closes an orbit, two fill a surface, a converging run never returns, and mini-batch noise fills the space it is given.

Theorem 2 (Fractal delay embedding prevalence; Sauer et al., 1991)**.** *Let $A \subset \mathbb{R}^n$ be compact with box-counting dimension d, invariant under a smooth diffeomorphism f defined on an open neighbourhood of A, and let $E > 2d$ be an integer. Assume that for each $p \leq E$ the set of periodic points of f of period p has box-counting dimension less than $p/2$, and that the linearisation of f at each such orbit has distinct eigenvalues. Then for almost every smooth ϕ, in the sense of prevalence, the delay map equation (4) is one-to-one on A and an immersion on every compact subset of a smooth manifold contained in A.*

The dimension in the condition $E > 2d$ is the box-counting dimension of the occupied set $\mathcal{A}_R$, which need not be an integer, so the criterion applies to the sets a training run occupies and not only to tori. Prevalence, in turn, is a measure-theoretic notion of "almost every" on an infinite-dimensional space.

Mini-batch sampling makes the update depend on a draw from the data as well as on the state, so neither theorem above applies as stated.

Theorem 3 (Delay embedding for forced systems; Stark, 1999, Stark et al., 2003)**.** *Let $\boldsymbol{s}_{t+1} = f(\boldsymbol{s}_t, w_t)$ with the drive w_t generated by a measure-preserving system on a space Ω, either deterministically or as an independent draw selecting among finitely many maps. Under genericity conditions on f and ϕ analogous to those of theorem 1, and for E exceeding twice the dimension of a fibre, the map sending a state to its delay vector equation (4) is an embedding* of each fibre M_ω, *that is, of the set of states compatible with one realisation ω of the drive.*

The guarantee holds fibre by fibre, whereas the sample available in practice is one trajectory, which lies in a different fibre at every step. No invariant measure on a fibre is being sampled, so the neighbour statistic below estimates nothing.

Theorem 4 (Maximum likelihood intrinsic dimension; Levina and Bickel, 2004, MacKay and Ghahramani, 2005)**.** *Let points be drawn from a smooth density supported on a d-dimensional manifold embedded in $\mathbb{R}^E$. Model the points falling within a small ball about $\boldsymbol{y}_i$ as a homogeneous Poisson process in the d-dimensional radial variable, with the density constant over the ball. Writing $r_j^{(i)}$ for the distance from $\boldsymbol{y}_i$ to its j-th nearest neighbour, the maximum likelihood estimate of d from the first m neighbours of that point is*

$$\hat{d}_m(\boldsymbol{y}_i)^{-1} \;=\; \frac{1}{m-1} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}. \tag{5}$$

Under this model the reciprocal is unbiased and the estimate itself is not.

The estimator is therefore a local statistic: it reports the exponent over the range of radii the m nearest neighbours span rather than the limit equation (1) takes. For $0 < \varepsilon_1 < \varepsilon_2$ that exponent at a point $\boldsymbol{c}$ is

$$d_R(\boldsymbol{c}; \varepsilon_1, \varepsilon_2) \;=\; \frac{\log \mu_R\big(B(\boldsymbol{c}, \varepsilon_2)\big) - \log \mu_R\big(B(\boldsymbol{c}, \varepsilon_1)\big)}{\log \varepsilon_2 - \log \varepsilon_1}. \tag{6}$$

Equation (3) pools it over the N points of the record. It agrees with $d_{\mathrm{act}}(R)$ when the same power of ε continues below ε_1. A phase of smaller amplitude than ε_1 is invisible to it however long the record. A record of N points resolves a dimension of at most about $2 \log_{10} N$ under the argument of Eckmann and Ruelle [1992], a bound appendix L finds does not explain the ceiling we measure.

B The estimator, its configuration and the observers

B.1 The estimator

Algorithm 1 is the estimator as it was run, with the departures from equation (3) that affect the values reported.

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ; embedding dimension E; neighbours m; Theiler exclusion W_T; dither σ

$z_t \leftarrow (x_t - \operatorname{mean} x)/\operatorname{std} x + \sigma\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1)$ ▷ breaks exact ties
$\boldsymbol{y}_t \leftarrow (z_t, z_{t-\tau}, \ldots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \ldots, n$
$S \leftarrow 0$; $N \leftarrow 0$; $n_{\text{dist}} \leftarrow 0$; $n_{\text{ratio}} \leftarrow 0$
for each $\boldsymbol{y}_i$ **do**
  $r_1^{(i)} \le \cdots \le r_m^{(i)} \leftarrow$ distances to the m nearest $\boldsymbol{y}_j$ with $|i-j| > W_T$
  $r_j^{(i)} \leftarrow \max(r_j^{(i)}, \varepsilon_{\text{dist}})$ for each j; count those clipped into n_{dist}
  $S_i \leftarrow \max\big(\sum_{j=1}^{m-1} \log(r_m^{(i)}/r_j^{(i)}),\ \varepsilon_{\text{ratio}}\big)$; count clips into n_{ratio}
  $S \leftarrow S + S_i; \quad N \leftarrow N + 1$
end for
mark the window degenerate if n_{dist} or n_{ratio} exceeds $1\,\%$ of its denominator
return $\big(N(m-1) - 1\big)/S$

The returned value is $(N(m-1)-1)/S$, the exactly unbiased pooled form, which differs from equation (3) negligibly at the sizes used here. *A window marked degenerate is returned rather than discarded.* No published cell of table 12 contains a marked window. *The returned value is never clamped to* E, so a divergent estimate is reported as divergent.

The Theiler exclusion is $W_T = \max\big((E-1)\tau,\ t_{\text{acf}}\big)$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples; appendix I.1 measures the effect of that cap. The floors $\varepsilon_{\text{dist}}$, $\varepsilon_{\text{ratio}}$ and the dither σ are 10^{-8}, 10^{-5} and 10^{-9}.

B.2 The observers

Table 1 defines the twelve observers of section 5.2.

Table 1: The scalar observers. $\boldsymbol{c}$ is the coordinate in the k-dimensional subspace and $\boldsymbol{\theta}_0$ the frozen head at initialisation; $\boldsymbol{L}$ is the matrix of probe logits, where the probe is a fixed held-out set drawn once before training; $\boldsymbol{g} = \partial\ell/\partial\boldsymbol{c}$. The projection directions $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{a}$, and the rotation $\boldsymbol{R}$, are drawn once from a fixed seed and never redrawn.

name	family	definition
instantaneous loss	loss	the loss on the current minibatch
full-batch loss	loss	the loss on the whole training set
probe loss	loss	cross-entropy on the probe set
parameter norm	norm	$(\lVert\boldsymbol{\theta}_0\rVert^2 + \lVert\boldsymbol{c}\rVert^2)^{1/2}$
subspace norm	norm	$\lVert\boldsymbol{c}\rVert_2$
function-space norm	norm	$\lVert\boldsymbol{L}\rVert_F$
gradient norm	gradient	$\lVert\boldsymbol{g}\rVert_2$
gradient projection	gradient	$\langle\boldsymbol{u}, \boldsymbol{g}\rangle$
fixed parameter projection	projection	$\langle\boldsymbol{v}, \boldsymbol{R}\boldsymbol{c}\rangle$
function-space projection	function	$\langle\boldsymbol{a}, \operatorname{vec}\boldsymbol{L}\rangle$
margin	function	mean probe logit gap, true class less best other
probe accuracy	function	accuracy on the probe set

B.3 Frozen configurations

One frozen configuration serves each range of the active dimension, each selected once on data withheld from every later experiment (table 2).

Table 2: The frozen configurations and their selection. Only the swept parameters were varied; the remainder were fixed before any estimate was computed. Settings common to both configurations are not repeated: $m = 20$ neighbours, a Theiler exclusion of at least the embedding span (appendix B.1). The neighbour count and the exclusion were swept in the eight-component grid, in which the two Theiler rules return bit-identical values, so the selection between them is arbitrary.

	eight-component	twenty-component
E_{max}	20	40
τ	4	16
window / stride	8 000 / 2 000	8 000 / 4 000
swept	E_{max}, τ, m, Theiler, window	E_{max}, τ
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–3	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.17	2.47
selection error, worst	2.03	3.35

The selection objective in each case is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection. The eight-component objective adds a penalty on the rank correlation and on the spread across seeds.

Each configuration lies at a grid boundary, at the top of every axis it was swept over, so a wider grid could move the level.

The selection error spans a factor of twelve on identical data (table 2), so the level is as much a choice as a measurement. Only the withheld numbers of section 5.2 demonstrate recovery.

The stride varies between experiments. Table 3 states the window geometry each experiment uses; none of them overrides a setting of the estimator. The constructed systems produced the recovery of section 5.2 at a stride of their own, so "at the frozen configuration" is true there of the estimator and not of the stride.

Table 3: The window and stride of each experiment, with their departures from the frozen configuration of table 2. Lengths are in samples of the log; the second stride applies to the longer records.

experiment	window	stride
frozen configuration	8 000	2 000
constructed systems (section 5)	8 000	3 000, or 3 666
training logs (section 7)	a third of the record	1 000

C Per-observer results and alternative statistics

Table 4: How accurately the estimator recovers the active dimension on each system, with the requirements of section 4 that each fails. Systems are ordered by increasing realism. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run's sliding windows; ρ is over ranks after a median over seeds. The oscillating matrix has no optimiser, so the zero-learning-rate check does not apply to it. The function-subspace row fails constructed truth at its three highest ranks, which reach an effective rank of about $0.86\,r$ against the $0.9\,r$ the check asks.

system	truth	range	MAE	ρ	requirements failed
oscillating matrix	constructed	1–8	0.27	1.00	*not applicable*
online linear regression	constructed	1–20	0.79	0.99	frozen configuration, zero learning rate
logistic regression	constructed	1–20	0.81	1.00	frozen configuration, zero learning rate
frozen nonlinear decoder	constructed	1–20	1.58	0.94	zero learning rate
perceptron in a k-subspace	constructed	1–20	0.92	0.99	zero learning rate
image data, parameter subspace	measured	1–8	0.90	1.00	**none**
image data, parameter subspace	measured	1–20	1.75	1.00	withheld ranks
image data, function subspace	measured	1–8	0.28	1.00	all of them

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.2, in the deterministic recurrent regime, scored against the measured effective rank rather than against r. Table 5 reconciles the three errors this paper quotes for that one measurement.

Table 5: Why three different errors are quoted for one measurement. Each row is the same eight-component recovery of section 5.2, on withheld ranks and withheld seeds in the deterministic recurrent regime, summarised in a different order. The order is stated wherever one of these figures appears.

aggregation	reported in	error
mean over (rank, seed, observer) runs	section 5.2, table 4	0.90
median over observers of each observer's error	table 6	0.48
median over seeds and observers at each rank, then scored	table 8	0.382

Table 6: The accuracy of each scalar observer, against the smoothness that might explain it. One row per observer, except that the parameter norm and the subspace norm share one, their errors agreeing to two decimals. The rank correlation is 1.00 for every observer, so it has no column. The last column records whether the roughness orders the withheld ranks perfectly.

observer	family	MAE	slope	roughness orders rank
probe loss	loss	0.60	0.82	no
function-space norm	norm	0.39	0.96	no
margin	function	0.46	0.83	no
parameter, subspace norm	norm	0.41	1.04	no
fixed parameter projection	projection	0.34	0.83	no
gradient projection	gradient	0.70	0.72	no
function-space projection	function	0.51	0.78	no
gradient norm	gradient	1.81	1.58	no
full-batch loss	loss	1.87	1.51	no
instantaneous loss	loss	0.61	0.73	no
probe accuracy	function	degenerate in every window		

Two observers are excluded from every aggregate. Probe accuracy is quantised, which collapses the delay vectors onto one another. The instantaneous mini-batch loss fails the zero-learning-rate check: frozen, it still climbs by more than five components across the withheld ranks and returns the estimate it returns when trained. The other eleven observers are exactly constant at zero learning rate, so no estimate is defined for them.

Nothing in a training log says which observer is the accurate one, so a reader can act only on the typical observer and not on the best row of table 6.

The highest rank correlation the roughness reaches over the four withheld ranks is 0.8, so smoothness alone accounts for no observer's agreement with the truth (figure 4).

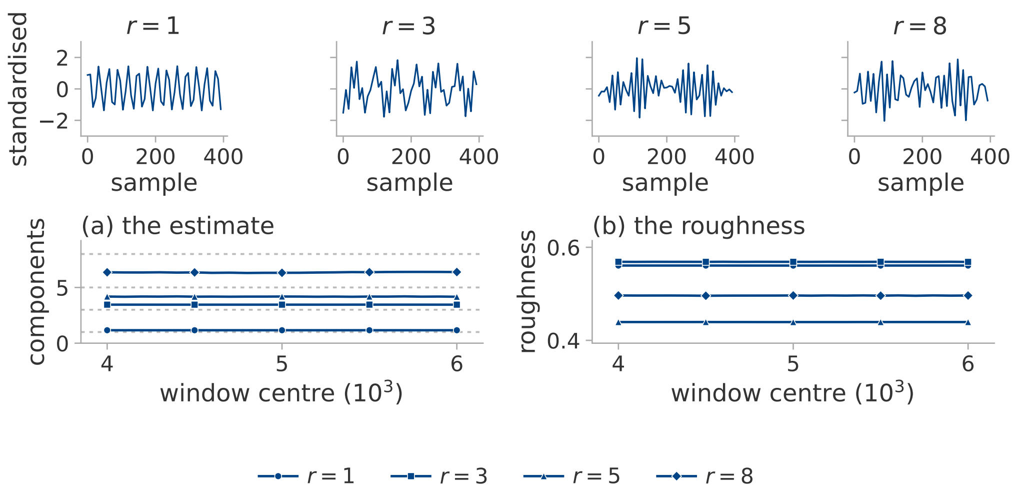

Figure 4: Whether four logs that look alike separate under the estimator. The image-data system of section 5.2 at four ranks and a seed the configuration was not selected on, through the parameter norm. Above, four hundred samples of each log. (a) The estimate over sliding windows, the measured effective rank dotted. (b) The roughness of the same windows.

Table 7: How far recovery extends at $k = 20$. Image data, median over three seeds and over the eight observers recorded there, which reports the typical observer rather than the best one; the truth is the measured effective rank. The last row is the spread of MG across seeds at fixed rank. Exploratory: this grid was scored with a smaller Theiler exclusion than section 4 asks (appendix I.1).

true r	1	2	4	6	8	10	12	14	17	20
MG	1.10	2.05	4.51	6.51	7.45	9.92	10.23	11.40	13.90	15.44
$\mathrm{PR}_{\mathrm{delay}}$	2.00	2.26	3.79	5.42	7.90	9.45	9.43	11.25	11.43	13.42
spectral PR	1.09	1.27	2.13	2.89	4.10	4.60	5.51	6.22	6.81	7.48
spread across seeds	0.87	0.51	1.35	1.81	2.64	3.59	2.81	2.07	3.41	4.24

Table 8: Whether any simpler statistic recovers the active dimension as well as the estimator does. All are computed on the image-data system of section 5.2. The two ranges are not aggregated alike: at $r \leq 8$ each statistic is scored on the withheld ranks alone, at $r \leq 20$ on all twenty, the selection ranks included, so only the comparison down a column is out of sample. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum.

	$r \leq 8$		$r \leq 20$	
statistic	MAE	ρ	MAE	ρ
MG †	0.382	1.00	1.75	0.997
LB †	0.40	1.00	1.52	0.997
TwoNN †	2.13	0.80	2.79	0.880
$\mathrm{PR}_{\mathrm{delay}}$	0.91	1.00	2.48	0.985
spectral PR, 256 bands	1.25	1.00	5.83	0.998
spectral PR, 1024 bands	1.08	1.00	4.78	0.979
spectral PR, native resolution	0.69	1.00	2.95	0.976
roughness	3.65	0.20	9.89	−0.379

Three qualifications attach to table 8. LB, the uncorrected per-point average, is more accurate than MG at $r \leq 20$: the correction of section 3.2 removes a positive bias, which on a saturating estimate was partly cancelling the saturation. The spectral participation ratio at 256 bands orders the ranks at $r \leq 20$ more faithfully than MG, at three times the error. The linear delay participation ratio needs no neighbour search at all, yet MG beats it by only a factor of about two at $r \leq 8$.

D Conditions of validity in detail

The delay lag. Figure 5 varies the lag τ alone. Below a tenth of a period every rank above one returns about 2.5 and the ordering is lost; between three tenths and a half of a period the estimate follows the rank; above one period it diverges.

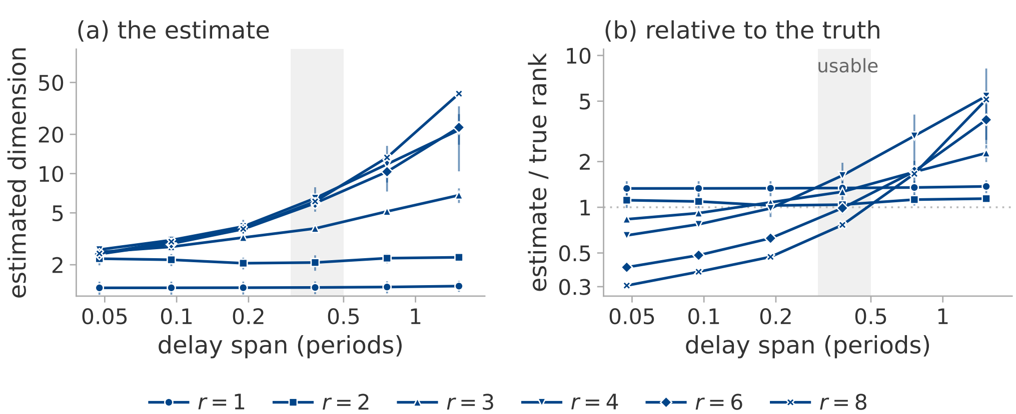

Figure 5: The delay lag, varied on one system: a fixed torus, two seeds, six lags spanning a twentieth of a period to one and a half periods. Marker shape is the true rank, the marker the median of the seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period.

Nuisance factors. The rotated-coordinate row of table 9 is scored on the fixed parameter projection alone, since the other eleven observers are invariant to a rotation by construction; it measures a property of the observer rather than of the estimator.

Table 9: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale for the others. The last column is a percentage of a real four-component change, which the change-detection experiment below observes as a fall of 3.96. A constant rescaling of the observer moves the estimate by 0.000 and is omitted.

factor	$\lvert\Delta\mathrm{MG}\rvert$	as % of a real four-component change
none (baseline)	0.39	10
learning rate halved	0.37	9
drive band up one octave	0.50	13
drive band down one octave	0.78	20
coordinates rotated	0.28	7
observer gain ramped $10\times$	1.53	38
state amplitude ramped $4\times$	1.57	39

Detection of a change. We run three consecutive segments $r_{\mathrm{high}} \to r_{\mathrm{low}} \to r_{\mathrm{high}}$, re-measuring the ground truth on each. In the deterministic recurrent regime a true fall of 4.00 components is observed as 3.96, is detected both downward and upward on over nine tenths of the runs, and recovers to within 0.05 of the level before it (table 10). Under stochastic excitation the same true change is observed as 0.09.

The rule declares a change when the estimate falls below the median of the pre-switch segment by three times the window-to-window scatter of that segment, for two consecutive windows.

Table 10: Whether a change of the active dimension is recovered at both levels and after it reverses. Three schedules $r_{\mathrm{high}} \to r_{\mathrm{low}} \to r_{\mathrm{high}}$, on the parameter norm. At $6 \to 2 \to 6$ the high level is over-reported by a component.

schedule	high	low	high again
$4 \to 1 \to 4$	4.06	0.84	4.06
$6 \to 2 \to 6$	7.00	2.17	6.95
$8 \to 3 \to 8$	8.32	3.34	8.28

Unequal drive amplitudes. Figure 6 gives the measurement of section 6.4 in full, observed through the squared Frobenius norm and estimated with the frozen eight-component configuration.

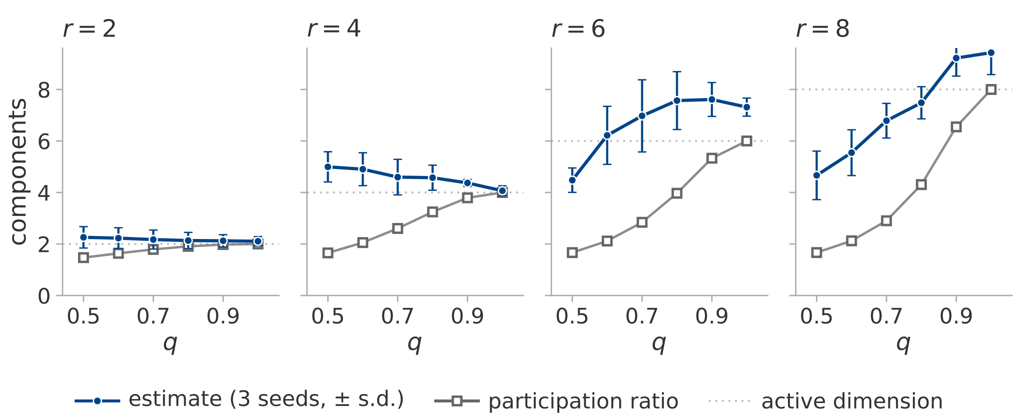

Figure 6: Whether the estimate follows the active dimension or the effective rank. Driving phase l at amplitude q^l holds the active dimension at r while the effective rank falls with q. The system of section 5.1, three seeds, one panel per r; the dotted line is r. Filled circles, the estimate, median over seeds, with standard deviation as bars; open squares, the measured effective rank, which matches its closed form to two decimals and so has no spread.

The estimate is flat in q at $r = 2$ and $r = 4$ while the effective rank declines, as intended, and declines itself at $r = 6$ and $r = 8$. The amplitudes show why: at $q = 0.5$ and $r = 8$ the weakest phase is driven at 0.008 of the strongest, far below the neighbourhood scale of the estimator, so the torus it resolves really does have fewer dimensions than the one that was built. The estimator therefore reports the number of phases it can resolve. That is neither r nor the effective rank; the three coincide only when the amplitudes are equal.

E System constructions

The image-data system of section 5.2. The data is partitioned into twelve fixed groups, the same partition at every rank. The loss weights of r of those groups are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j. These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\mathrm{pinv}(\Phi)Q$ with Q an orthonormal basis of $\mathrm{range}(\Phi)$ orthogonalises them. The per-phase scalar response gain of the linearised update is then divided out.

Table 11 measures how many independent directions each construction excites. The transient and the plain mini-batch rows report an outcome rather than a setting: a converging trajectory is a curve whatever r is and the rank of mini-batch noise is fixed by the per-example gradient covariance of the data, so projecting it onto r directions recovers only part of the intended rank and the estimates in that regime are scored against the measured value.

Table 11: Whether each construction excites as many independent directions as intended. Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.2. Every estimate on this system is scored against these values, never against r.

regime	$r = 1$	2	4	6	8	excitation
`qp`	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast drive
`qp_slow`	1.00	2.00	4.00	5.99	7.98	the same at a training-log timescale
`qp_nopre`	1.00	2.00	3.73	5.98	7.77	`qp` without the preconditioner
`mixed`	1.00	2.00	4.00	6.00	7.99	`qp` plus noise in the same directions
`noise`	1.00	2.00	4.00	5.99	7.99	injected rank-r gradient noise
`noise_nopre`	1.09	1.98	3.71	5.24	6.75	the same without the preconditioner
`batch_proj`	1.00	1.74	3.08	4.19	5.38	real mini-batch noise projected to rank r
`batch`	6.58	6.58	6.56	6.60	6.56	plain mini-batch descent
`gd`	1.04	1.05	1.04	1.04	1.06	a decaying transient
`qp`, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	the zero-learning-rate check

The functional rank of the backbone is 10 at every rank, seed and regime, a property of the frozen backbone rather than of the excitation.

The function-space construction. A local adapter is built around a trained network of widths 64, 16 and 10. Its function-space Jacobian is QR-whitened, so that the first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it. The functional, trajectory-covariance and update-covariance ranks are measured rather than assumed. Eight of the sixteen withheld cells fall short of the $0.9r$ effective rank the construction requires, at $0.87r$ in the median. Table 4 therefore marks this variant as failing every requirement.

The system at twenty available directions. The same construction at $k = 20$ reproduces the nominal rank over $r = 1, \ldots, 20$ in the recurrent regime; every other regime behaves as it does at $k = 10$.

F Diagnostics and figures for the grokking application

Table 12 gives the diagnostics behind section 7.1.

Table 12: Whether the estimate is a dimension in either grokking setting. The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is a median over the run's sliding windows. Seven of the seventeen runs of table 19 are shown; figure 7 plots all seventeen. A "no" is a run that had not generalised when training ended, which is a censored observation.

run	generalises	MG	ρ_{ident}	roughness	PR_{delay}	crossings
transformer, full data	yes	13.32	1.53	0.097	1.03	204
transformer, reduced data, s0	no	4.70	1.59	0.067	1.50	161
transformer, reduced data, s2	no	4.42	1.55	0.139	4.04	153
transformer, no weight decay	no	15.60	1.76	< 0.001	1.00	2
perceptron, $n + m$	yes	15.08	1.42	0.001	1.00	2
perceptron, $n \cdot m$	yes	14.97	1.39	0.001	1.00	2
perceptron, $n^3 + nm^2 + m$	no	15.41	1.51	< 0.001	1.00	2

Figure 7 places every training log of section 7 in the plane of the identifiability ratio and the trend-crossing count, against the region in which every accurate case of section 6 fell. No run lies in that region: every one is unstable in the embedding dimension, at ratios well above the 1.10 the region allows. On the other axis the transformer and perceptron runs fail in opposite directions, since the perceptron series are monotone.

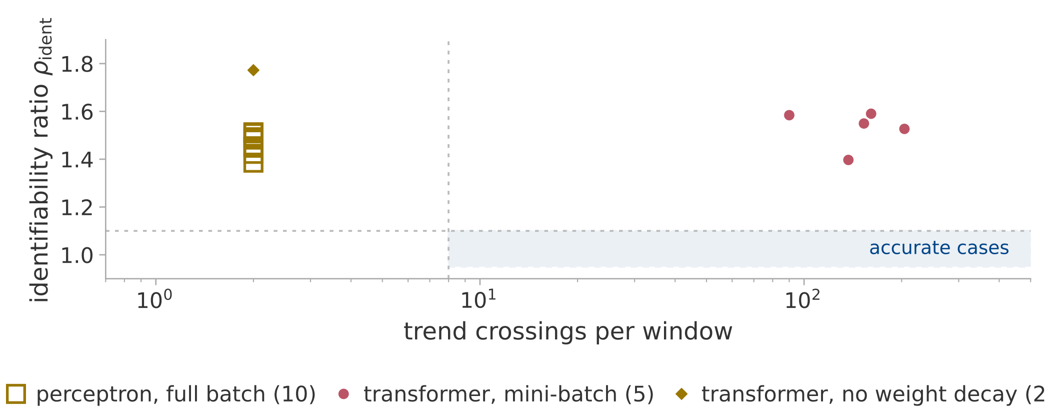

Figure 7: Whether any grokking log lies where the estimate was accurate. Every log of section 7 in the plane of the identifiability ratio and the trend-crossing count, the transformer runs observed through the parameter norm and the perceptron runs through the training loss. Colour is the regime of section 3.3, marker shape the setting. The shaded rectangle is where every accurate case of section 6 fell. It is a record of those cases, not a test.

Figure 8 shows the separation of section 7.1 and the mechanism behind it: the medians separate every pair, though the ranges overlap.

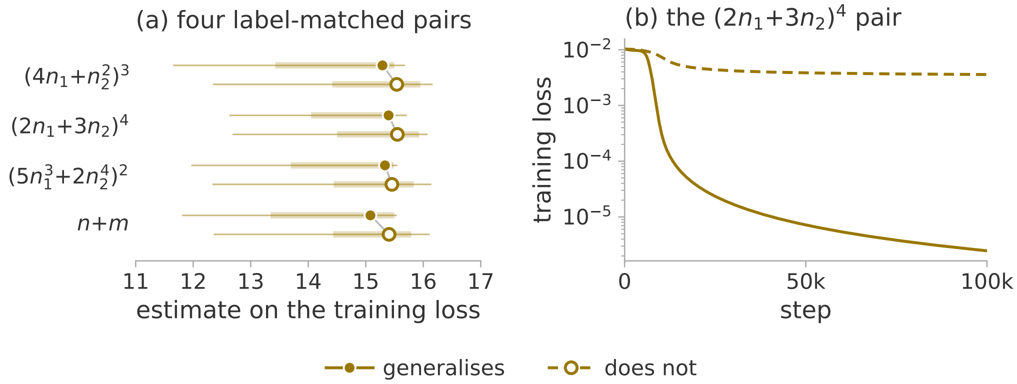

Figure 8: Whether the estimate tells apart two targets that differ by one monomial, only one of them representable by the architecture. (a) The pairs of section 7.1, on the training loss. Marker, median over the sliding windows; thick bar, interquartile range; thin line, full range. All runs are transient, hence one colour; fill and dash mark generalisation. (b) One pair in full: the representable target drives the loss orders of magnitude lower.

G The direct trajectory measurement

G.1 The trajectory sketch

The direct measurement of section 7.2 needs the trajectory itself: the whole parameter vector at each of two to three thousand logged steps. Every logged step is therefore compressed to $\mathbb{R}^{1024}$ by a CountSketch [Charikar et al., 2002, Weinberger et al., 2009], a sparse random projection drawn twice from two independent hash families so that the two copies can be compared. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length, without which a signal taken from the logits could not be told from a measurement of the weight decay its scale is coupled to.

We do not verify a spectral guarantee for the compression. The two hash families supply a direct estimate of the error instead: every reported value is the mean of the two projections, with the disagreement between them recorded beside it. That disagreement is a per cent of the value at the median and under nine per cent at its worst, smaller than every effect section 7.2 reports.

The sketch is accurate on synthetic trajectories of known rank. It agrees to within 0.11 components throughout (table 13). Nothing above rank ten is tested, however, whereas the plateaus in figure 2 are higher.

It agrees with exact arithmetic on a real run. On the perceptron architecture the participation ratio computed on all the raw parameters agrees with the sketched value to three decimals.

Recording the trajectory does not perturb training. The log is required to be bit-identical with and without the recording, over four hundred steps at reduced probe size. One global random stream supplies the split, the initialisation and the mini-batch order in turn, so a single stray draw changes the initial weights and can destroy grokking.

Table 13: The cost of the compression. The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in $145\,500$ dimensions. The difference between the first two columns is the cost of a sixty-sample participation ratio and is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch − uncompressed
1	1.00	1.00	0.00
2	1.99	1.99	0.00
5	4.56	4.54	−0.02
10	8.44	8.33	−0.11

Windows are sixty logged rows, each labelled by its centre, so a feature at a given step is located to within half a window. The statistic reported per window is $\mathrm{PR}^{\mathrm{det}}$, the detrended effective rank of section 3.1. The total displacement is recorded beside it, since a rank can fall simply because the trajectory has stopped moving.

G.2 The effective-rank collapse, run by run

The plateau of table 14 is the median over the 4 000 steps before the generalisation step, the dip the minimum within $\pm 4\,000$ steps of it, and "at" the position of that minimum relative to it. The two runs without weight decay have no generalisation step, so each is measured over the window of the run it matches. In parameter space the post-dip maximum exceeds the plateau by an order of magnitude in the modular runs. In function space it returns to about the plateau and exceeds it in two of the four runs.

Table 14: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The post-dip maximum is the largest value reached afterwards: the sense of "re-expands" in section 7.2. The two rows without weight decay in each block are measured in the aligned window; their deepest fall anywhere in their own run is 3.15 and 7.06 in function space and 1.66 and 2.37 in parameter space, the comparison section 7.2 also reports.

space	run	plateau	dip	depth	at	post-dip max
function	mod_wd1	24.24	3.80	6.38	135	23.99
	mod_wd1_s43	33.02	3.86	8.55	455	27.80
	mod_wd1_s44	27.16	3.40	7.99	515	30.82
	s5_wd1	21.04	2.62	8.03	1 413	23.53
	mod_wd0	5.50	3.43	1.60	—	—
	s5_wd0	11.87	9.09	1.31	—	—
parameter	mod_wd1	3.01	1.09	2.76	1 035	36.12
	mod_wd1_s43	2.57	1.15	2.24	1 155	37.34
	mod_wd1_s44	2.71	1.13	2.39	1 315	37.44
	s5_wd1	7.14	1.39	5.14	1 513	12.46
	mod_wd0	29.93	27.70	1.08	—	—
	s5_wd0	5.57	4.49	1.24	—	—

H The resolution of a windowed statistic

H.1 The full-batch setting

Repeating the direct measurement on the perceptron setting gives a limit of the measurement rather than a negative finding. Under full-batch gradient descent every window of every run has a participation ratio between 1.00 and 1.36 on the statistics the comparison uses.

A deterministic update makes the trajectory a smooth curve. Over a 600-step window such a curve is straight to within the measurement precision, so the statistic reports the curvature of the trajectory at the scale the window sets, where there is almost none. The statistic has no range in which a collapse could appear.

A range of window lengths settles the question (table 15). Two runs of 150 000 steps, a generalising one and its label-matched partner, are measured over windows spanning 600 to 118 000 optimiser steps, each holding the same sixty samples spread further apart as the window lengthens, so that the bound on the statistic and its noise floor stay where the published measurement put them. The statistic does leave its floor, at a scale of about 10^4 steps, more than an order of magnitude above the published window. The absence of a feature at the shorter window was therefore a limit of resolution rather than a fact about the run.

The feature that appears there separates nothing. One localised maximum appears, of the same size in both runs (table 15); function space agrees, with more range and no more separation. Nor can a window this long place that maximum to better than the interval between memorisation and generalisation. Beyond 30 000 steps the ordering even reverses and the non-generalising run moves further, because the window is then a large fraction of a run whose weight norm is still climbing, so that the statistic tracks that drift. The full-batch trajectory therefore carries no signature of generalisation in its effective rank at any window length between 600 and $118\,000$ steps: the collapse of section 7.2 belongs to the regularised stochastic setting.

Table 15: Whether the full-batch statistic has any range at longer windows. Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window, sixty samples per window at every length.

	generalises		control, never generalises	
window (steps)	median	max	median	max
600	1.002	1.035	1.001	1.011
2 950	1.066	1.258	1.003	1.087
5 900	1.027	1.534	1.005	1.360
11 800	1.010	1.951	1.020	1.826
29 500	1.006	2.047	1.112	1.735
59 000	1.025	1.551	1.345	1.912
118 000	1.231	1.626	2.009	2.271

Introducing mini-batches confirms this. The same statistic then ranges over more than a decade while the generalisation step moves by a single logged step. Any claim based on a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

H.2 Declines of the estimate that are not generalisation

A decline of the estimate is not specific to generalisation. In a long run at $p = 211$ with no weight decay, which memorises early and ends at chance validation accuracy, a decline is detected tens of thousands of steps after memorisation. The ramped-gain experiment of section 6.3 supplies a mechanism for it that needs no dimension at all: the parameter norm of this run roughly triples, acting on the estimate through the observer. No trajectory was stored for that run, so nothing here says whether it simplified; at the frozen window and in these settings the estimate on a log is in any case neither a dimension nor an effective rank.

Section 7.3 does not make the decline specific either. The estimate on a log does decline where the trajectory does, in four runs and in neither of the other two, but at a window some seventy times shorter than the one used here, under a configuration chosen for that window.

H.3 Resolution of a windowed estimate on a training log

The change section 7.1 leaves open is testable on the seven extended runs. The frozen configuration takes a window of a third of each record, $39\,990$ optimiser steps, advanced by $10\,000$ steps at a time. That gives nine windows per run, localising a feature only to $\pm 19\,995$ steps. The runs therefore cannot be aligned on their generalisation steps as figure 2 aligns them: no two share an offset, so no mean over runs can be formed.

The one run with a window centre near its generalisation step *rises* by 2.93 components over the following stride and by 2.71 over the one after. Those are the only two of the fifty-six stride-to-stride changes to exceed the 1.53-component floor a ramped gain alone produces (section 6.3). The other two generalising runs stay inside it throughout, as do all four that never generalise (figure 9). A rise is not the fall section 7.3 reports, and at $\pm 19\,995$ steps neither could be assigned to the transition.

As in appendix H.1, the statistic has no range in which a feature confined to the transition could appear. Testing the claim requires a window set in units of the width of the transition, which for these runs is hundreds of steps.

The matched-window re-run. Section 7.3 performs that re-run on the six runs of figure 2 rather than the seven of this appendix: the question needs a paired comparison and only those six had their trajectory stored. Each window is 60 samples, centred on the midpoint of a window of section 7.2. The significance of the fall is assessed against the surrogates of section 7.3 [Theiler et al., 1992, Schreiber and Schmitz, 1996].

The frozen configuration does not fit. At 60 samples its delay span of 76 exceeds the window. The grid crosses $E_{\max} \in \{4, 6, 10\}$, $\tau \in \{1, 2, 4\}$, $m \in \{5, 20\}$ and both Theiler rules, giving thirty-six cells, of which twenty-six are long enough to return a value. Twenty-four of those separate the four generalising runs from the other two. The headline cell is named by the rule of section 7.3, which does not choose the best cell: a neighbouring cell separates the runs more sharply.

The diagnostics do not survive the shortening. The identifiability ratio needs an embedding of $2E_{\max}$, so at the headline cell the Theiler exclusion leaves too few usable points to compute one. Where the ratio can be computed at all, the two Theiler rules disagree about whether it passes; it settles nothing.

Linear detrending removes the effect. Repeating every cell on a linearly detrended window leaves none that separates the generalising runs from the rest. Detrending a sketch and detrending a scalar are not comparable: on a 1024-dimensional sketch a per-coordinate linear detrend removes one direction out of many, whereas on a scalar series it removes most of the series.

The observer matters. On the training loss rather than the parameter norm, twelve of the twenty-six cells separate them, though the surrogate comparison passes in the S_5 run only.

The two runs without weight decay have almost no dynamic range. At this window the baseline belongs to the configuration rather than to the run: on a standardised straight line equation (3) returns about 2.77, close to the pre-transition level of every run. The parameter-norm log of the modular run is straight to under a per cent of its own spread, so its estimate spans three thousandths of a component against a range three orders of magnitude larger in the generalising runs; the S_5 run is a weaker case of the same thing. Their D and quantile columns therefore record an observer that offered the estimator no local structure to respond to. They are not dimensional negatives. The four generalising runs are untouched, and so is the surrogate comparison, in which each run is measured only against surrogates of itself.

Table 16: The fall of the log estimate at generalisation. The matched-window re-run of section 7.3 on the parameter-norm log, at the headline cell $E_{\max} = 10$, $\tau = 1$, $m = 20$, embedding-span Theiler. D is the median estimate over the three thousand steps before the transition divided by the minimum over the window that brackets it; the quantile is the rank of D among the same statistic at every window centre after memorisation; the floor is where that minimum falls, relative to t_{gen}. The p columns are the fraction of surrogates, at three smoothing lengths, whose D is at least the observed one.

run	D	quantile	floor	p_{101}	p_{201}	p_{401}
modular, seed 42	2.96	0.99	−65	0.025	0.025	0.025
modular, seed 43	1.98	0.93	255	0.050	0.050	0.025
modular, seed 44	2.14	0.92	315	0.050	0.075	0.025
S_5 composition	2.12	0.89	1 313	0.050	0.050	0.050
modular, no decay	1.00	0.46	−865	0.950	0.950	0.925
S_5, no decay	1.04	0.38	1 613	0.975	0.975	1.000

Put through the same comparison, the linear participation ratio responds less than the estimator and the roughness, as section 7.3 reports, far more.

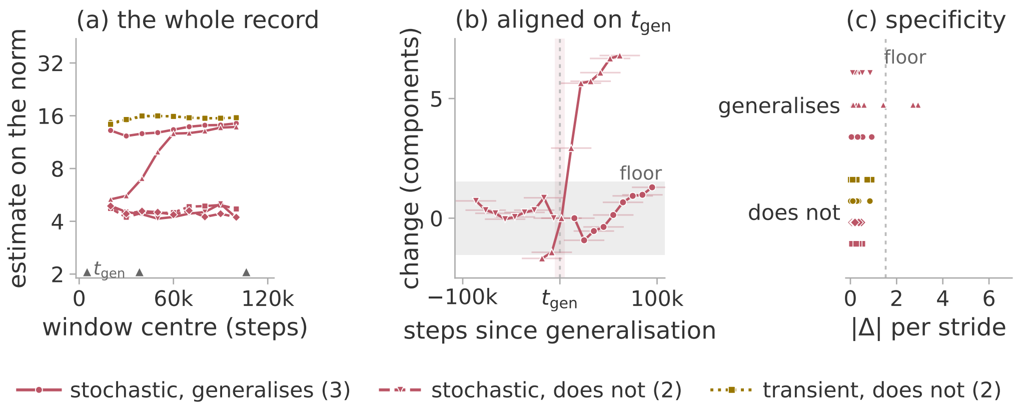

Figure 9: How coarsely a frozen-window estimate localises a feature. The windowed estimate on the parameter norm for the seven extended runs, under the frozen configuration of section 7.1, plotted at the window centres. Colour is the regime of section 3.3. (a) Every window, with the generalisation steps marked on the axis. (b) The generalising runs aligned on their own generalisation step. (c) Every stride-to-stride change, one sub-row per run. The grey band in (b) and (c) is the floor a ramped gain alone produces (section 6.3).

I The Theiler exclusion

I.1 The exclusion cap at the twenty-component configuration

The rule of appendix B.1 asks for at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap; the implementation caps that at 150 samples.

At the eight-component configuration the embedding span is 76, so the cap binds only where the autocorrelation time exceeds 150. That never happens on the constructed systems, whose autocorrelation time is a few samples, but it happens on five of the seven parameter-norm records of section 7. Those five were scored at $W_T = 150$ rather than at the value the rule asks for, in the direction that inflates the estimate. At the twenty-component configuration the span is 624 and the cap binds unconditionally. It is why section 5.2 marks that result exploratory.

Table 17 measures the cost of that departure.

Table 17: The cost of the exclusion cap. The twenty-component system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured $\mathrm{PR}^{\mathrm{pos}}$ and is not comparable with table 4, which covers more observers.

r	measured $\mathrm{PR}^{\mathrm{pos}}$	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	1.74	1.81
5	5.00	4.16	5.15
8	8.00	6.28	6.71
11	11.00	9.54	10.26
14	14.00	9.81	9.37
17	17.00	14.40	14.19
20	19.99	15.92	16.60
mean absolute error		2.16	1.89
Spearman ρ		1.00	0.96

The ordering of the ranks, all that section 5.2 claims at this range, is nearly unaffected, a single inversion apart. The mean absolute error falls when the protocol is enforced, so the cap was not making the reported result look better than it is. The saturation above about six components is present at both settings and is therefore not an artefact of the exclusion.

The identifiability ratio inherits a second effect of the cap: its numerator is computed at $2E_{\max}$ with τ held fixed, so that delay span exceeds the cap while the denominator's does not.

The five records the cap already binds on escape it, since both estimates there sit at 150. It bites on the remaining two and, more consequentially, on the constructed systems of section 5: the reference band of section 6.4 was itself measured with 76 against 150. A ratio built from two differently excluded estimates is not the quantity section 3.4 describes.

I.2 The Theiler exclusion and the dimension of a transient

Section 3.3 claims that recurrence is a requirement of this protocol and not of the estimator. The claim is testable on the excitation cases of section 5.2 by varying the Theiler exclusion W_T alone, from zero up to the value the frozen rule gives (table 18).

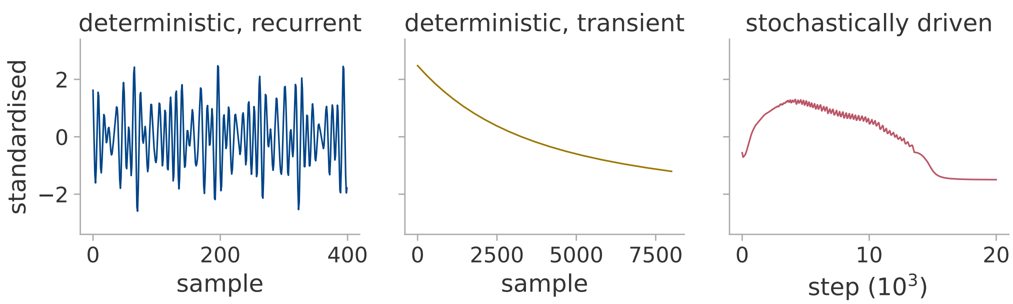

Figure 10: The three regimes as raw scalar logs. Each panel is the parameter norm, standardised to zero mean and unit variance as in Algorithm 1. Left, the constrained head of section 5.2 under a fast quasiperiodic drive, recurrent on a torus. Middle, the same system under plain gradient descent, transient over its whole record. Right, a transformer grokking modular addition, stochastically driven.

Table 18: The estimate against the Theiler exclusion, median over observers, seeds and windows. The transient's estimand is 1: at $W_T = 0$ the estimator returns it, whereas the exclusion removes it. The fast recurrent case is unchanged under the same variation. The last column removes the cap of 150 and applies the autocorrelation rule, some 1 600 samples here.

W_T	0	2	5	20	100	150	uncapped
recurrent, fast	4.83	4.82	4.82	4.82	4.80	4.80	4.80
recurrent, slow	2.06	3.02	3.22	3.27	3.27	3.28	3.27
transient	1.20	1.71	2.38	5.51	20.50	29.22	173.19

The value at $W_T = 0$ is the right one. On a uniformly sampled curve the nearest neighbours of a point are its immediate temporal neighbours, for which equation (3) returns 1.199; the transient returns that value in every cell. The excess over unity is the lattice bias of a deterministic sampling.

The exclusion removes the near neighbours, not the neighbours. The estimator fills its twenty slots at every exclusion; only the distances change. Before any exclusion the nearest neighbours of a point lie thousands of samples away under the fast quasiperiodic drive, but six samples away on the transient (figure 10). Excluding 150 samples therefore costs the fast drive almost none of its close pairs and the transient all of them (figure 11). The distances the transient retains are then nearly equal, their log-ratio sum collapses, and the reciprocal in equation (3) diverges. An exponential decay with no optimiser and no data behind it reproduces both ends.

The asymmetry is general; the value 29 *is not.* Under the fast drive $W_T = 0$ already returns r, because that case is not oversampled; it is the slow drive that the exclusion repairs, its nearest neighbours lying some seven hundred samples away. A transient also need not have dimension one: a decaying *oscillation* correctly returns 2.36 at every exclusion.

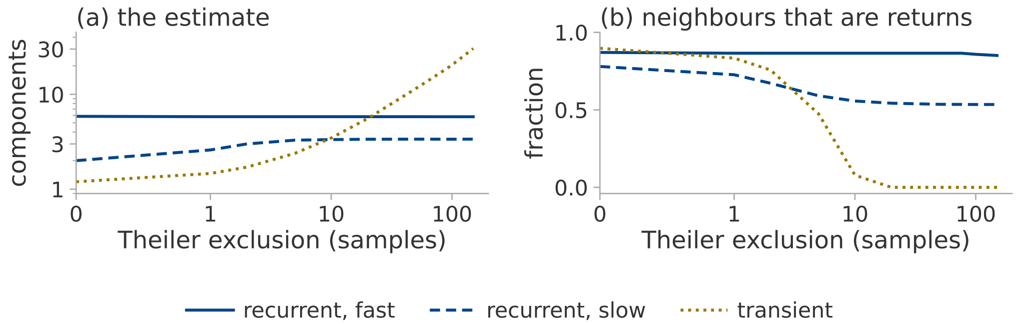

Figure 11: Why the exclusion sets the value on a transient and leaves a torus alone. The excitations of table 18, median over ranks, seeds and windows. In (b) a neighbour counts as a return when it is no further away than the nearest neighbours an unexcluded search finds, so a fall to zero leaves the estimator nothing but distant points.

A recurrence-based diagnostic. The ratio $\hat{d}_{MG}(W_T{=}150)/\hat{d}_{MG}(W_T{=}20)$ is unity under both recurrent drives and above five on the transient, so it tests recurrence in the reconstructed space where the trend-crossing count tests only non-monotonicity of the series: the gap section 3.4 leaves open, which appendix K shows to matter on real runs. The principled version is older than either: a space–time separation plot [Provenzale et al., 1992] shows directly how neighbour distance depends on separation in time, and the exclusion should be chosen from one.

J The training runs

Table 19 lists every training run the paper draws on. **T** is the one-layer transformer of Nanda et al. [2023]: width 128, four heads, no layer normalisation, about $227\,000$ parameters, trained in double precision with AdamW. **P** is the two-layer quadratic perceptron of Gromov [2023]: width 500, $145\,500$ parameters at $p = 97$, trained by full-batch gradient descent with no regularisation.

Nanda et al. [2023] train at full batch and we use mini-batches. In the perceptron setting we take the tasks and the representability criterion of Doshi et al. [2024] but not their optimiser, running plain full-batch gradient descent on mean squared error instead. The S_5 task [Power et al., 2022] and its weight decay lie outside the configuration of Nanda et al. [2023]. The two S_5 runs also step the optimiser twice per batch, so their effective learning rate and decay are about double the nominal ones.

Definition of memorisation and generalisation. The memorisation step is the first logged step at which the training accuracy reaches $95\,\%$; the generalisation step is the first at which the validation accuracy does. Neither requires the threshold to be held thereafter. The extended reruns of section 7.1 use the stricter rule instead, requiring the threshold to hold over a sustained block.

Table 19: Training runs used in this paper. α is the training fraction, "final" the validation accuracy at the end of training. A dash in the generalisation column marks a run that had not generalised when training ended; such runs are censored observations, not established negatives.

run	task	arch	wd	α	steps	mem.	gen.	final	used in
Trajectory sketch attached									
`mod_wd1`	mod. add. p=113	T	1.0	0.30	20k	1470	13 660	1.00	section 7.2
`mod_wd1_s43`	mod. add. p=113	T	1.0	0.30	20k	1580	6640	1.00	section 7.2
`mod_wd1_s44`	mod. add. p=113	T	1.0	0.30	20k	1410	3680	1.00	section 7.2
`s5_wd1`	S_5 composition	T	0.2	0.50	15k	1280	6685	1.00	section 7.2
`mod_wd0`	mod. add. p=113	T	0.0	0.30	20k	630	—	0.03	section 7.2
`s5_wd0`	S_5 composition	T	0.0	0.50	15k	1015	—	0.01	section 7.2
Extended reruns, 120 000 *steps*									
`grokpos_s0`	mod. add. p=113	T	1.0	0.30	120k	1450	5070	1.00	section 7.1
`lowdata15_s0`	mod. add. p=113	T	1.0	0.15	120k	620	—	0.58	section 7.1
`lowdata15_s1`	mod. add. p=113	T	1.0	0.15	120k	630	106 380	1.00	section 7.1
`lowdata15_s2`	mod. add. p=113	T	1.0	0.15	120k	660	—	0.01	section 7.1
`lowdata20_s0`	mod. add. p=113	T	1.0	0.20	120k	800	38 380	1.00	section 7.1
`wd0_s0`	mod. add. p=113	T	0.0	0.30	120k	610	—	0.09	section 7.1
`wd0_s1`	mod. add. p=113	T	0.0	0.30	120k	630	—	0.07	section 7.1
`p211_wd0`	mod. add. p=211	T	0.0	0.16	200k	1800	—	0.04	section 7.1
Perceptron, $p = 97$*, full batch, no regularisation*									
`a_add`	$n + m$	P	0	0.5	100k	9230	11 760	1.00	section 7.1
`a_mul`	$n \cdot m$	P	0	0.5	100k	9100	11 560	1.00	section 7.1
`a_sub`	$n - m$	P	0	0.5	100k	9110	11 540	1.00	recorded only
`a_sq_sum`	$n^2 + m^2$	P	0	0.5	100k	8440	10 470	1.00	recorded only
`a_sum_sq`	$(n + m)^2$	P	0	0.5	46k	7350	8190	1.00	recorded only
`x_mix_quad`	mixed quadratic	P	0	0.5	100k	9600	—	0.01	figure 7
`x_no_grok`	$n^3 + nm^2 + m$	P	0	0.5	100k	9860	—	0.01	section 7.1
`g_p1`	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7470	12 290	1.00	section 7.1
`g_p1x`	$+ n_1 n_2$	P	0	0.5	100k	10 140	—	0.01	section 7.1
`g_p2`	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6470	7680	1.00	section 7.1
`g_p2x`	$- n_1^2$	P	0	0.5	100k	9730	—	0.02	section 7.1
`g_p3`	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5970	6600	1.00	section 7.1
`g_p3x`	$- n_2$	P	0	0.5	100k	9420	—	0.74	section 7.1

Four of the perceptron runs are repeated with the trajectory sketch attached, for the measurement of appendix H.1: `a_add` and `x_no_grok` from the table above, together with the label-matched pair `g_p1` and `g_p1x`. Each is run at full batch and again with mini-batches of 512. The full-batch runs reproduce the memorisation and generalisation steps of the rows above exactly and the mini-batch runs to within six logged steps, so appendix H.1 compares two noise levels on matched learning.

`g_p3x` ends far above the majority-class baseline of $1.3\,\%$ while being provably outside the class of functions the architecture can represent.

K Descent at the edge of stability

Descent at the edge of stability [Cohen et al., 2021] is gradient descent at a fixed rate: the top Hessian eigenvalue sharpens until it reaches $2/\eta$ and then stays there, while the loss descends over long stretches and not over short ones. Such a run is deterministic and is not a plain transient, so an undriven trajectory could satisfy the conditions of section 3.3.

Setting. We run full-batch descent without weight decay on the quadratic perceptron of appendix J (table 20). The log is written at *every* optimiser step, because the oscillation is the two-cycle of the unstable mode and its period is a few steps. The sharpness is measured by power iteration on Hessian-vector products, taking the largest *algebraic* eigenvalue, since the Hessian is indefinite early in training.

Table 20: Oscillation and recurrence in undriven descent at the edge of stability. Full-batch descent by learning rate, both seeds. The stability ratio is $\eta\lambda_{\max}/2$ over the second half of the run; *rises* is the fraction of post-transition steps on which the loss increases; *return* is the nearest neighbour surviving a 150-sample exclusion divided by the distance the trajectory travels during that exclusion, for which a two-frequency torus scores 0.008.

η	stability ratio	rises	return	outcome
1×10^5	$0.239, 0.238$	0.000	1.007	monotone
3×10^5	$0.758, 0.745$	0.000	$1.007, 1.007$	monotone
1×10^6	$0.969, 0.971$	$0.495, 0.490$	$0.104, 0.079$	at the edge
1.5×10^6	$0.974, 0.970$	$0.500, 0.500$	$1.012, 1.011$	at the edge
2×10^6	$0.977, 0.978$	$0.500, 0.500$	$0.120, 1.013$	at the edge
2.5×10^6	—, 0.988	—, 0.500	—, 0.034	one seed diverges
2.8×10^6	0.992, —	0.503, —	0.022, —	one seed diverges
3×10^6	—	—	—	diverges

Oscillation is not recurrence. Eight runs reach the edge and all eight oscillate, the loss rising on one step in two, but only five of them return close to states already passed through. The return column of table 20 separates those two groups by a factor of ten with nothing in between.

Performance of the diagnostics. The band of section 3.4 rejects all eight runs at the edge, on the training loss and on the parameter norm alike, whether or not they recur, ρ_{ident} lying far outside it in both groups while the trend-crossing count, which tests non-monotonicity and not recurrence, passes every one of them. Nor does the estimate separate the runs that return from those that do not: their ranges on the training loss overlap.

Dependence on the logging stride. Written at the stride every other full-batch log here uses, most of the phenomenon disappears (figure 12): over these eight runs the median trend-crossing count falls from 71 to 6. Whether a log looks like a transient is thus in part a property of how often it was written.

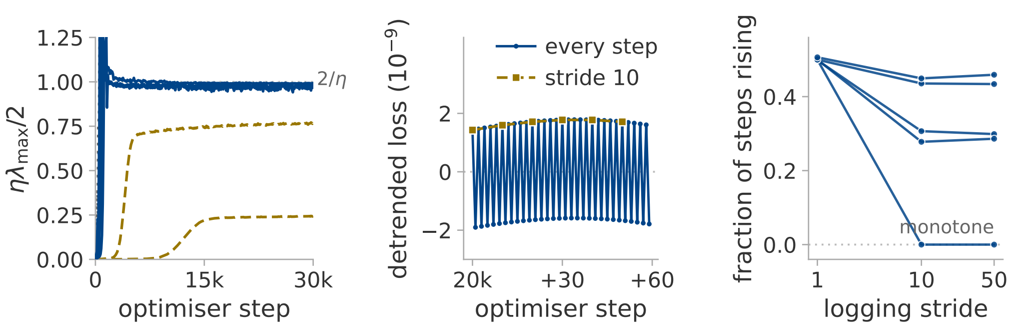

Figure 12: Whether full-batch descent reaches the edge of stability. (a) The stability ratio along training, one seed per rate. (b) Sixty consecutive steps at one rate at the edge, against the same series at a stride of ten: the two-cycle is removed by that stride rather than blurred by it. (c) The fraction of steps on which the loss rises, against the logging stride.

L The ceiling: embedding dimension against record length

Two explanations were available for the ceiling of section 5.2: the embedding condition $E > 2d$ and the finite-record bound $2\log_{10} N$ [Eckmann and Ruelle, 1992]. Each is insensitive to the variable the other names, so we sweep $E_{\max}$ at a fixed record length and the record length at fixed $E_{\max}$, over $r \in \{2 \dots 20\}$ with everything else frozen (table 21). Each record length is a *prefix* of one longer record rather than a resampling of a fixed span, leaving the delay lag its meaning in periods. Every trajectory is bit-identical to the published one over their common prefix.

Table 21: Whether the ceiling follows the embedding condition or the finite-record bound. Two sweeps, three seeds per cell. *Tracking* is the largest r at which the median estimate stays within one component of the truth; *level* is the median estimate at $r = 20$; *slope* is $\mathrm{d}\hat{d}_{MG}/\mathrm{d}r$ over $r \geq 8$, which is 1 if the estimator still tracks the rank and 0 if it has stopped.

$E_{\max}$ at $N = 8000$	10	14	20	28	40	56	
tracking	6.5	7.3	8.1	10.3	10.9	11.2	
$E_{\max}/2$ *predicts*	5	7	10	14	20	28	
level at $r = 20$	6.9	8.1	9.8	11.3	13.2	14.9	
slope	0.08	0.12	0.20	0.29	0.41	0.54	
N at $E_{\max} = 20$	10^3	$2{\cdot}10^3$	$4{\cdot}10^3$	$8{\cdot}10^3$	$1.6{\cdot}10^4$	$3.2{\cdot}10^4$	$6.4{\cdot}10^4$
tracking	6.7	8.3	8.5	8.1	8.9	9.5	9.5
$2\log_{10} N$ *predicts*	6.0	6.6	7.2	7.8	8.4	9.0	9.6
level at $r = 20$	9.0	9.3	9.4	9.8	9.9	10.2	10.5
slope	0.22	0.17	0.17	0.20	0.19	0.19	0.22

Outcome. The level at high rank is set by the embedding: a sixty-fourfold increase in the record adds 1.5 components, whereas a sixfold increase in $E_{\max}$ adds eight. The tracking ceiling follows neither variable. It never passes 11.2, yet at a quarter of the record the rest of the study uses it still tracks to 8.3, where $2\log_{10} N$ allows only 6.6. A bound set by the record length cannot produce that, so Eckmann and Ruelle [1992] is not the explanation here. Nor is $E > 2d$ in the form we used it: the ceiling rises by a tenth of a component per unit of $E_{\max}$ rather than by a half: $E_{\max}/2$ happens to be right near $E_{\max} = 14$ and is wrong at both ends.

Fitting the level at high rank separates the candidates (table 22): neither hypothesis fits, nor does their pointwise minimum, whereas a form logarithmic in both variables fits an order of magnitude more closely than the spread of the data itself.

Table 22: Whether either candidate explanation of the ceiling accounts for the measured level. Candidate forms fitted at $r \geq 16$ over every measured $(E_{\max}, N)$ cell; the spread of the fitted quantity itself is 2.00 components, so a form with a larger error than that describes nothing.

form	root-mean-square error
$E_{\max}/2$, the embedding condition	8.73
$2\log_{10} N$, the finite-record bound	3.68
the pointwise minimum of the two	3.70
$8.61\log_{10} E_{\max} + 0.90\log_{10} N - 5.31$	0.19

A purely linear statistic reproduces the whole dependence. Across the $E_{\max}$ sweep $\mathrm{PR}_{\mathrm{delay}}$ rises over the same range as MG; across the record sweep it is flat; and a single logarithm in $E_{\max}$ fits it more tightly than any form fits anything else in this appendix. Most of the ceiling is therefore the number of components the delay window resolves *linearly*.

The unexplained tracking limit. The ceiling lies near eleven components for no reason we can give. It does not follow the measured effective rank, which equals r throughout, nor does any form we fitted account for it.

The identifiability ratio of section 3.4 does register it. At $r = 20$ the ratio stays above the band at every record length, so a longer record never makes an unidentifiable estimate identifiable; it falls into the band as $E_{\max}$ grows, at about the rank where tracking stops.

M Two controls separating the estimator from roughness

The number of components changes and the roughness does not. Both systems in this control are a sum of four sinusoids. In the first the four frequencies are rationally independent, so the path lies on a four-dimensional torus and $d_{\mathrm{act}} = 4$. In the second they are harmonics of one irrational base frequency: a single phase fixes all four sinusoids at once, the path is a circle, and $d_{\mathrm{act}} = 1$. The base frequency of the second system is then set by bisection so that the two have the *same* roughness, to a part in 10^{15}. A scheduled run passes between the two through ramped envelopes, leaving unscored every window a ramp reaches into (figure 13).

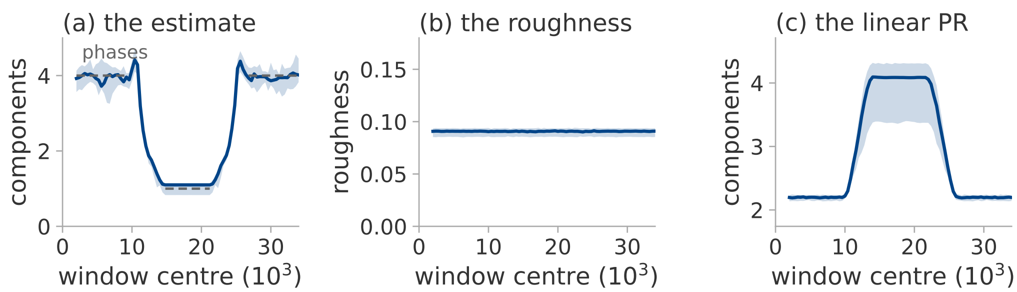

Figure 13: A change in the number of independent phases, tracked as it happens. The harmonic control of this appendix, scheduled from four phases to one and back; median over eight seeds, band their range. Dashed in (a), the number of phases, drawn only where no ramp reaches into the window. The linear statistic of (c) moves in the opposite direction.

The roughness changes and the number of components does not. Five monotone observers $g(x) = x + ax^p$ are applied to one four-phase system. Each is invertible, so it may bend the recorded set but cannot add or remove a coordinate. Each also changes the local scale $g'(x)$ and with it the spacing of consecutive samples.

Table 23: Both controls, median over eight seeds. Above the rule, two systems matched in roughness and differing in the number of independent phases. Below it, one four-phase system under five invertible maps. $\mathrm{PR}_{\mathrm{delay}}$ is the linear participation ratio of the delay covariance, included because it follows the roughness rather than the active dimension in both halves.

	d_{act}	estimate	roughness	$\mathrm{PR}_{\mathrm{delay}}$
four independent phases	4	3.92	0.091	2.20
one phase, four harmonics	1	1.11	0.091	4.10
identity	4	3.92	0.091	2.20
$x + 0.25x^3$	4	3.96	0.098	2.32
$x + x^3$	4	3.96	0.110	2.58
$x + x^5$	4	4.01	0.146	3.61
$x + 0.1x^7$	4	4.02	0.166	4.28

Both criteria were fixed before the runs, the second of them as a tenth of a component about four. Table 23 shows both hold.

Both controls run on a constructed system whose phases are known, so they say nothing about a training log.

Two findings count against the estimator. The linear participation ratio is higher on the circle than on the four-torus (table 23), so anyone treating it as a count of phases would order the two backwards. On the circle the estimate is also not geometric: every neighbour surviving the exclusion there is a sample from another pass around the circle. The local speed of the curve multiplies all of a point's neighbour distances alike; speed cancels from the log-ratios of equation (3). The residue depends on the arithmetic of the base frequency: two of the eight seeds drew a frequency whose multiples cluster into near-duplicates; their estimate is reproducible from the fractional parts of those multiples alone, with no signal and no neighbour search.

N Compute cost and code availability

Compute cost. Everything here runs in a few hours on one consumer GPU and one desktop CPU. The largest GPU item is the edge-of-stability experiment of appendix K, which takes half an hour on a single T4; the analysis is the more expensive half and runs entirely on CPU, a few core-hours per experiment.

Cost of the trajectory sketch. At each logged step the sketch keeps 4096 floats against the full parameter vectors of appendix J: a run whose raw trajectory would occupy 88 megabytes occupies 2.5. We train the same configuration with and without the sketch attached, alternating the order so that warm-up cannot favour either. At a logging stride of ten steps the sketch adds about six per cent to the training time. `sketch_cost.py` records the measurement.

Code availability. All code, configurations and result files behind every number and figure in this paper will be released publicly on acceptance. The figures regenerate from the committed result files with a single command.